

Portfolio Site Build

Design and Build Your Professional Web Presence

■ LEARNING OUTCOME

By the end of this module you will design and build a complete professional portfolio site with semantic HTML, responsive CSS, smooth animations, contact form, and deploy it live – a site that works as your 24/7 job application.

■ Skills You Will Gain

- Design a compelling portfolio layout with clear sections
- Write semantic HTML5 with proper meta tags for SEO
- Implement responsive design with CSS Grid and Flexbox
- Add smooth scroll animations and hover effects
- Build a working contact form with Formspree
- Optimise images and achieve Lighthouse score above 90

■ Your portfolio is your most important project. 70% of hiring decisions start with a web presence check. A polished portfolio with live projects gets 5x more callbacks than one without a URL to share.

SECTION 1

Portfolio Architecture and Content

Section	Purpose	Key Content
Hero	First impression -- 5 seconds to hook	Name, title, one-line value prop, two CTAs (View Work / Contact Me)
About	Build trust and personality	Photo, 2-3 paragraphs, skills icons, CV download button
Skills	Quick scan for recruiters	Tech icons grouped by category (Frontend, Backend, Tools)
Projects	Proof of ability	3-5 best projects with screenshot, tech tags, live demo + code links
Experience	Career timeline (if any)	Company, role, dates, 3 bullet achievements per role
Contact	Convert visitor to conversation	Form with name/email/message, social links, email address
Footer	Navigation and attribution	Quick links, social media, built-with credit

SECTION 2

HTML Structure

```
<!DOCTYPE html> <html lang='en'> <head> <meta charset='UTF-8'> <meta name='viewport'
content='width=device-width, initial-scale=1.0'> <!-- SEO meta tags --> <meta name='description'
content='Alice Sharma -- Full Stack Developer specialising in React and Node.js'> <meta
name='keywords' content='Full Stack Developer, React, Node.js, Mumbai'> <!-- Open Graph for social
sharing --> <meta property='og:title' content='Alice Sharma | Full Stack Developer'> <meta
property='og:description' content='Building fast, accessible web apps'> <meta property='og:image'
content='https://alicecode.dev/og-image.jpg'> <meta property='og:url'
content='https://alicecode.dev'> <meta name='twitter:card' content='summary_large_image'>
<title>Alice Sharma | Full Stack Developer</title> <link rel='canonical'
href='https://alicecode.dev'> <link rel='stylesheet' href='style.css'> <link rel='preconnect'
href='https://fonts.googleapis.com'> <link rel='preload' as='image' href='hero-bg.webp'> <link
rel='icon' href='/favicon.ico'> </head> <body> <!-- Skip link for accessibility --> <a href='#main'
class='skip-link'>Skip to main content</a> <header class='site-header' role='banner'> <nav
class='nav' aria-label='Main navigation'> <a href='#top' class='nav__logo'>Alice Sharma</a> <ul
class='nav__list' role='list'> <li><a href='#about'>About</a></li> <li><a
href='#projects'>Projects</a></li> <li><a href='#contact'>Contact</a></li> </ul> <button
class='nav__toggle' aria-label='Toggle menu' aria-expanded='false'>
<span></span><span></span><span></span><span></span> </button> </nav> </header> <main id='main'> <section
class='hero' id='top' aria-label='Introduction'> <div class='hero__content'> <p
class='hero__greeting'>Hi, I am</p> <h1 class='hero__name'>Alice Sharma</h1> <p
class='hero__title'>Full Stack Developer</p> <p class='hero__bio'>I build fast, accessible, and
beautiful web applications with React and Node.js.</p> <div class='hero__cta'> <a href='#projects'
class='btn btn--primary'>View My Work</a> <a href='#contact' class='btn btn--outline'>Let's
Talk</a> </div> </div> </section> <section class='projects' id='projects'
aria-labelledby='projects-heading'> <h2 id='projects-heading'>Featured Projects</h2> <div
class='projects__grid'> <article class='project-card'> <img src='proj1.webp' alt='Screenshot of
e-commerce dashboard showing sales charts' width='600' height='400' loading='lazy'> <div
```

```

class='project-card__body'> <h3 class='project-card__title'>E-Commerce Dashboard</h3> <p>Real-time
sales analytics with React and Chart.js.</p> <ul class='tags' aria-label='Technologies used'>
<li>React</li><li>Node.js</li><li>MongoDB</li> </ul> <div class='project-card__links'> <a
href='https://demo.alicecode.dev' target='_blank' rel='noopener' aria-label='Live demo of
E-Commerce Dashboard'>Live Demo</a> <a href='https://github.com/alice/ecommerce' target='_blank'
rel='noopener' aria-label='GitHub code for E-Commerce Dashboard'>GitHub</a> </div> </div>
</article> </div> </section> </main> <footer class='site-footer'> <p>&copy; <time
datetime='2024'>2024</time> Alice Sharma. Built with HTML & CSS.</p> </footer> <script defer
src='app.js'></script> </body> </html>

```

SECTION 3

CSS Design System

```

/* ===== DESIGN TOKENS ===== */ :root { /* Brand colours */ --color-primary: #10B981;
--color-primary-dark: #059669; --color-bg: #0B1D18; --color-surface: #132018; --color-text:
#F0FDF4; --color-text-muted: #9CA3AF; --color-border: #1F4033; /* Typography */ --font-sans:
'Inter', system-ui, sans-serif; --font-heading: 'Poppins', sans-serif; /* Type scale */ --fs-sm:
clamp(.875rem, 1.5vw, 1rem); --fs-base: clamp(1rem, 2vw, 1.125rem); --fs-lg: clamp(1.25rem, 3vw,
1.75rem); --fs-xl: clamp(2rem, 5vw, 3.5rem); --fs-hero: clamp(2.5rem, 8vw, 5.5rem); /* Spacing */
--sp-xs: .5rem; --sp-sm: 1rem; --sp-md: 1.5rem; --sp-lg: 3rem; --sp-xl: 6rem; /* Effects */
--shadow-sm: 0 1px 3px rgba(0,0,0,.4); --shadow-lg: 0 8px 32px rgba(0,0,0,.5); --radius: 12px;
--transition: .3s cubic-bezier(.4,0,.2,1); } /* ===== RESET ===== */ *, *::before, *::after {
box-sizing: border-box; margin: 0; padding: 0; } html { scroll-behavior: smooth; } body {
font-family: var(--font-sans); background: var(--color-bg); color: var(--color-text); line-height:
1.6; } img { max-width: 100%; height: auto; display: block; } /* ===== HERO ===== */ .hero {
min-height: 100svh; display: grid; place-items: center; background: radial-gradient(ellipse at 60%
40%, rgba(16,185,129,.15) 0%, transparent 70%); text-align: center; padding: var(--sp-lg)
var(--sp-sm); } .hero__name { font-size:var(--fs-hero); font-family:var(--font-heading);
line-height:1; } .hero__title { font-size:var(--fs-lg); color:var(--color-primary); } /* =====
PROJECTS GRID ===== */ .projects__grid { display: grid; grid-template-columns: repeat(auto-fill,
minmax(min(320px,100%),1fr)); gap: var(--sp-md); } .project-card { background:
var(--color-surface); border-radius: var(--radius); overflow: hidden; border: 1px solid
var(--color-border); transition: transform var(--transition), box-shadow var(--transition); }
.project-card:hover { transform: translateY(-6px); box-shadow: var(--shadow-lg); } /* =====
ANIMATIONS ===== */ .fade-in-up { opacity: 0; transform: translateY(30px); } @media
(prefers-reduced-motion: no-preference) { .fade-in-up.visible { animation: fadeInUp .6s ease-out
forwards; } } @keyframes fadeInUp { to { opacity: 1; transform: translateY(0); } }
.project-card:nth-child(2) { animation-delay: .1s; } .project-card:nth-child(3) { animation-delay:
.2s; } /* ===== CONTACT FORM ===== */ /* Use Formspree for free backend-less form handling */
form[action*='formspree'] input, form[action*='formspree'] textarea { width: 100%; padding: .75rem
1rem; background: var(--color-surface); border: 1px solid var(--color-border); border-radius: 8px;
color: var(--color-text); font-family: inherit; } form[action*='formspree'] input:focus,
form[action*='formspree'] textarea:focus { border-color: var(--color-primary); outline: none;
box-shadow: 0 0 0 3px rgba(16,185,129,.2); }

```

SECTION 4

JavaScript Interactions

```

// Scroll animations with IntersectionObserver const observer = new IntersectionObserver((entries)
=> { entries.forEach(entry => { if (entry.isIntersecting) { entry.target.classList.add('visible');
observer.unobserve(entry.target); // stop observing after animate } }); }, { threshold: 0.1 });
document.querySelectorAll('.fade-in-up').forEach(el => observer.observe(el)); // Mobile navigation
toggle const toggle = document.querySelector('.nav__toggle'); const navList =
document.querySelector('.nav__list'); toggle.addEventListener('click', () => { const expanded =
toggle.getAttribute('aria-expanded') === 'true'; toggle.setAttribute('aria-expanded', !expanded);
navList.classList.toggle('open'); }); // Active nav link on scroll const sections =
document.querySelectorAll('section[id]'); const navLinks = document.querySelectorAll('.nav__list
a'); const scrollSpy = new IntersectionObserver((entries) => { entries.forEach(entry => { if
(entry.isIntersecting) { navLinks.forEach(a => a.removeAttribute('aria-current')); const active =
document.querySelector(`.nav__list a[href="#${entry.target.id}"]`); if (active)
active.setAttribute('aria-current', 'page'); } }); }, { threshold: 0.5 }); sections.forEach(s =>

```

```
scrollSpy.observe(s)); // Project filter
const filterBtns = document.querySelectorAll('[data-filter]');
const cards = document.querySelectorAll('.project-card');
filterBtns.forEach(btn => {
  btn.addEventListener('click', () => {
    const filter = btn.dataset.filter;
    filterBtns.forEach(b => b.classList.remove('active'));
    btn.classList.add('active');
    cards.forEach(card => {
      const tags = card.dataset.tags || '';
      card.style.display = filter === 'all' || tags.includes(filter) ? '' : 'none';
    });
  });
});
```

■ PRACTICE Q & A

Q1. What is the Open Graph protocol and why does it matter for your portfolio?

A: OG meta tags (`og:title`, `og:description`, `og:image`) control how your site appears when shared on social media (LinkedIn, Twitter, WhatsApp). A great OG image with your name and title makes shares look professional and drives more clicks.

Q2. Why should you use WebP format for project screenshots?

A: WebP is 30-50% smaller than JPEG/PNG at the same quality. Smaller images = faster page load = better Lighthouse score = better SEO ranking. Convert PNGs to WebP with: `npx sharp-cli input.png -o output.webp`.

Q3. What is a Formspree and why use it over a backend?

A: Formspree is a free form backend service -- point form action to their URL and they email you submissions. No server, no cost, no maintenance. Perfect for a portfolio contact form. Upgrade to a custom backend only when you need the data in a database.

Q4. What Lighthouse score should your portfolio have?

A: Aim for 90+ on all four: Performance, Accessibility, Best Practices, SEO. Run in Chrome DevTools (F12 > Lighthouse). Fix every red/orange issue. A 90+ score proves you care about code quality to every developer who views source.

Q5. What is the skip link and why is it required?

A: A visually hidden link at the top that becomes visible on keyboard focus, allowing users to skip past the navigation directly to main content. Without it, keyboard and screen reader users must tab through every nav item on every page -- a critical accessibility failure.

■ Portfolio Build Cheat Sheet	
<code>og:image</code> meta tag	Social sharing preview image
<code>preload</code> critical images	<code>link rel=preload as=image</code>
<code>loading='lazy'</code>	Lazy-load below-fold images
WebP format	30-50% smaller than JPEG/PNG
<code>clamp(2rem, 5vw, 4rem)</code>	Fluid responsive font size
<code>grid auto-fill minmax</code>	Responsive project grid
<code>IntersectionObserver</code>	Scroll animations efficiently
<code>prefers-reduced-motion</code>	Respect OS animation setting
<code>aria-expanded</code> on toggle	Accessible mobile nav
<code>formspree.io</code>	Free contact form backend
<code>skip-link</code>	Accessibility keyboard navigation
Lighthouse 90+	Run before every launch